Dr. KATHIRVEL ALAGESAN

Max Planck Unit for the Science of Pathogens (MPUSP), Berlin, Germany
Email: alagesan@mpusp.mpg.de
kathiralagesan.github.io | ORCiD: 0000-0002-7596-5558

PROFILE & CORE EXPERTISE

Mass spectrometry research leader with **15+ years of experience in LC-MS analytical technologies**. Currently leading the Proteomics Research Platform at MPUSP, balancing strategic and operational leadership of a state-of-the-art MS facility with an independent scientific focus on **systems glycobiology**. Dedicated to enabling high-impact multidisciplinary research through the development of robust, FAIR-oriented analytical workflows and quality frameworks.

Core-Facility Leadership: Operational budget management • scientific user consultation • resource & capacity planning • MS instrumentation & project management • SOP development • personnel supervision.

Mass Spectrometry & Proteomics: LC-MS/MS • DDA/DIA/PRM • targeted & discovery proteomics • glycoproteomics • glycomics • phosphoproteomics • RNA-Protein XL-MS • PTM crosstalk • automated sample processing • glycopeptide synthesis.

Informatics, Quality & Infrastructure: Automated QC & instrument monitoring • FAIR analytical workflows • quantitative & statistical analysis • Open Science data standards • cross-institutional research infrastructure.

PROFESSIONAL EXPERIENCE

Project Leader in Proteomics **July 2022 – Present**

Proteomics Research Platform
Max Planck Unit for the Science of Pathogens, Berlin, Germany

- Lead the **Proteomics Research Platform**, providing end-to-end LC-MS-based analytical services and scientific support to **>30 researchers** across MPUSP and collaborating institutes **(200 registered project submissions, 25 external collaborations since 2021).**
- **Direct platform operations, resource planning, and quality management**, including MS instrumentation, maintenance, and service contracts; managing consumables, workflow capacity, Standard Operating Procedures (SOPs), user procedures, and quality control processes.
- **Oversee and forecast an annual operational budget**, evaluating and approving routine operational expenditures and service requirements while coordinating major strategic and higher-value decisions with the Director of MPUSP.
- Provide **pre-experimental scientific consultation**, advising researchers on experimental design, sample size, controls, sample preparation, and quantitative strategies to ensure robust, fit-for-purpose proteomics experiments.
- Develop and implement **high-throughput, project-specific proteomics and bioinformatics workflows**, including LC-MS/MS acquisition (DDA, DIA, and PRM), data processing, statistical analysis, enrichment and network analysis, and data visualization.
- Develop and evaluate advanced analytical strategies for **microbial and host–pathogen proteomics and glycobiology**, including system-wide characterization of low-abundance proteins, bacterial secretomes and small proteins, **glycoproteins**, phosphopeptides, and other post-translational modifications.
- Develop **reproducible, FAIR-oriented proteomics workflows, quality-control systems, and analytical tools** to improve data quality, standardization, and transparency, including **Karuppar**, an automated LC-MS/MS quality control and monitoring system for continuous assessment of instrument performance.
- **Supervise and train technical staff**, contribute to **staff recruitment and interview processes**.
- Supervise and co-supervise **master's practical courses and research internships,** and provide training and scientific mentoring to researchers and students in MS-based proteomics.

Post-Doctoral Research Fellow **Sep 2020 – June 2022**

Charpentier Lab, Max Planck Unit for the Science of Pathogens, Berlin, Germany

***Theme**: Role of glycan in host-pathogen interaction*

***Mentor:** Emmanuelle Charpentier*

- Established the **Glycoproteomics research theme** to investigate host-pathogen interaction.
- **Took charge of the Proteomics Research Platform in May 2021.**
- Developed analytical tools and techniques to study bacterial post-translational modifications.
- Advised postdoctoral researchers and PhD students on proteomics strategy, experimental design, LC-MS analysis, data interpretation, manuscript preparation, and scientific communication.

Post-Doctoral Research Fellow **Mar 2017 – Aug 2020**

Institute for Biomedicine and Glycomics (formerly Institute for Glycomics), Griffith University, Australia

***Theme**: Synthetic translational glycoproteomics*

***Mentors:** Daniel Kolarich, Mark von Itzstein*

- **Actively involved in planning and establishment of an advanced mass spectrometry facility**, including operating, troubleshooting, and maintaining the Ultimate 3000 nLC and Thermo Orbitrap Fusion, Bruker rapifleX MALDI-TOF mass spectrometers.
- Established the *'Glycomics & Glycoproteomics in Cancer and Evolution'* research group together with Prof. Daniel Kolarich.
- **Established the glycopeptide synthesis lab** and pursued individual research on synthetic translational glycoproteomics and clinical proteomics/glycoproteomics.
- Developed **MS methods and analytical workflows** for glycopeptide characterization, including a chemo-enzymatic workflow for diversification of *N*-glycopeptides.
- Developed new tools for automated glycan analysis from clinical samples.
- Secured internal and competitive research funding as **Principal and Associate Investigator** and established industry collaborations with **MassDynamics, HILICON, and RESYN** to support technology development and translational applications.

RESEARCH INFRASTRUCTURE, STANDARDS & SCIENTIFIC SERVICE

1. **Max Planck MS Core Unit Network** — active participant in the planning phase, contributing to discussions on coordination, infrastructure, and development of mass spectrometry core facilities within the Max Planck Institutes.
2. **Interlinking Berlin-Brandenburg Microbiome Research Expertise** — active participant in cross-institutional research networking and infrastructure development led by Sofia Kirke Forslund (MDC, Berlin)
3. **MIRAGE (Minimum Information Required for A Glycomics Experiment)** — invited participant and speaker at the Beilstein-Institut meeting focused on improving standards, reporting, and data quality in glycomics research.
4. **Glycoconjugate Exchange Format** — invited participant and speaker at the Beilstein-Institut meeting addressing standards and exchange of glycoscience data.
5. **Immunoglobulin G (IgG) glycan analysis - Brainstorm meeting:** Participated in scientific planning and expert discussions organized by Prof. Gordon Lauc.
6. **Expert panel member for re-accreditation of a postgraduate university program**, appointed by the Agency for Science and Higher Education, Croatia.

LEADERSHIP, MANAGEMENT & PROFESSIONAL DEVELOPMENT

1. **EMBO Laboratory Management Course for Group Leaders** — training in laboratory leadership, personnel management and research-group management.
2. **Introduction to Teaching in Higher Education** — higher-education teaching, supervision and curriculum development.
3. **Genetic Engineering Safety Regulations (GenTSV)**: Completed training for Project Leaders and Biosafety Officers; basic course (December 2022) and refresher course (June 2025).
4. **Open Science Ambassador, Max Planck Society** — promoting Open Science practices, transparency, reproducibility, and collaborative research within the Max Planck Society.

SCIENTIFIC & INSTITUTIONAL LEADERSHIP

1. **Chair**, 2024 MPUSP-HU Retreat on '**Regulatory Mechanisms in Bacteria**'.
2. **Organizer** of the Berlin Seminar Series in Microbial Sciences (2021-2025).
3. **Co-organizer** of the Berlin Microbiology Club (2021-2025).
4. **Co-chair** of the **2023 Gordon Research Seminar (GRS)** on **Glycobiology.**
5. **Event Organization –**2021 MPUSP Symposium & 2nd Griffith ECR Cross-Institute Symposium 2020.
6. **Member of the organizing committee** - 2022 & 2025 MPUSP **Scientific Advisory Board (SAB)**.
7. **Discussion leader –** 'Glycans in health and disease' session at **2019 GRS on Glycobiology**.

OUTREACH & COMMUNITY ENGAGEMENT

1. Participated in Long Night of Sciences – '**Science Slam - Of Humans and Microbes**'
2. **Glycomics Champions Program (GCP)** – one of the members to promote research to a broader community, including students, researchers from other institutions, philanthropists, and Government officials.
3. **Student member of the Teaching Development Group** at the Department of Biochemistry, Oulu University.

RESEARCH GRANTS

Principal Investigator

- **Griffith University New Researcher Grant** (2018) '*Development of a novel platform for synthesis of glycopeptide conjugate vaccine*' ***(AU$16,000).***

Associate Investigator

- **Research, Development and Innovation Grant** (2020) *'Vaccine for Control of Flystrike'* ***(AU$35,000****).*
- **Mizutani Foundation for Glycoscience** (2019) *'Identifying the human bisected N-acetylglucosamine modifying glycosyltransferase'* ***(US$25,000).***

Competitive Fellowship (2012-2015)

- **Beilstein doctoral fellowship** awarded by 'Beilstein-Institut', Germany ***(Euro 59,400)***

PEER REVIEW & SCIENTIFIC EVALUATION

Grant & Proposal Review

- **Expert Reviewer, GlycoMIP**, National Science Foundation, Materials Innovation Platforms, USA (since 2023).
- **Expert Reviewer, Dutch Research Council** (NWO), Netherlands (since 2025).

Journal & Academic Peer Review

- **Regular reviewer** for *Analytical Chemistry, MethodsX, Talanta, Glycoconjugate Journal, Analyst, mSphere, Proteomics,* and *Journal of Proteome Research*.
- **Evaluator**, PhD Thesis (The University of Queensland, Australia)

SELECTED PUBLICATIONS

Total record: 27 publications & book chapters. Total citations:>990 | h-index: 15.

Full list available at the end and on PubMed & Google Scholar.

Selected publications below highlight contributions most relevant to analytical technologies, microbial proteomics & host–pathogen biology.

Proteomics, Glycoproteomics & Analytical Technologies

1. **Alagesan K*** and Charpentier E (2023) "Systems-Wide Site-Specific Analysis of Glycoproteins" In K. Gevaert (Ed.), ***Mass Spectrometry-Based Proteomics*****,** 2718, 151–165. New York, NY: Springer US.
2. **Alagesan K**, Hoffmann M, Rapp E and Kolarich D (2020) "Glycoproteomics Technologies in Glycobiotechnology" In E. Rapp and U. Reichl (Eds.), ***Advances in Glycobiotechnology***, 175, 413–434. Cham: Springer International Publishing.
3. Kawahara R, Chernykh A, **Alagesan K**, Bern M, Cao W, Chalkley RJ, Cheng K, Choo MS, Edwards N, Goldman R, Hoffmann M, Hu Y, Huang Y, Kim JY, Kletter D, Liquet B, Liu M, … Thaysen-Andersen M (2021) "Community evaluation of glycoproteomics informatics solutions reveals high-performance search strategies for serum glycopeptide analysis" ***Nat Methods*****,** 18(11), 1304–1316.
4. De Leoz MLA, Duewer DL, Fung A, …. **Alagesan K**,… Stein SE (2020) "NIST Interlaboratory Study on Glycosylation Analysis of Monoclonal Antibodies: Comparison of Results from Diverse Analytical Methods" ***Molecular & Cellular Proteomics*****,** 19(1), 11–30.
5. **Alagesan K**, Hinneburg H, Seeberger PH, Silva DV and Kolarich D (2019) "Glycan size and attachment site location affect electron transfer dissociation (ETD) fragmentation and automated glycopeptide identification" ***Glycoconj J*****,** 36(6), 487–493.
6. **Alagesan K**, Khilji SK and Kolarich D (2017) "It is all about the solvent: on the importance of the mobile phase for ZIC-HILIC glycopeptide enrichment" ***Anal Bioanal Chem***, 409(2), 529–538.
7. Campbell MP, Nguyen-Khuong T, Hayes CA, Flowers SA, **Alagesan K**, Kolarich D, Packer NH and Karlsson NG (2014) "Validation of the curation pipeline of UniCarb-DB: Building a global glycan reference MS/MS repository" ***Biochimica et Biophysica Acta (BBA) - Proteins and Proteomics***, 1844(1), 108–116.

Microbial Proteomics & Host–Pathogen Biology

8. Rubinić M, Yu Y, **Alagesan K,** and Iatsenko I (2026) "A Pore-Forming Toxin Monalysin Contributes to Infection-Induced Suppression of Defecation in Female Drosophila" ***PLOS Pathog. Accepted.*** bioRxiv.
9. Cornejo FA, Driller K, Ahmed-Begrich R, Schmidt K, Jahn M, Shanmuganathan V, Hahnke K, Kondrot F, Wulff TF, Rämisch S, **Alagesan K**, Charpentier E, Turgay K (2026) "Stress-induced ribosome degradation in *Bacillus subtilis* is mediated by the RNase Y-specificity complex" ***Nat. Commun***. 17(1):4886.
10. Krause K, Franch Arroyo S, Ugolini M, Kueck T, Sullivan TJ, Gálvez EJC, Muenzner M, Goosmann C, Brinkmann V, Frese CK, **Alagesan K**, Vierbuchen T, Heine H, Resch U, Sander LE and Charpentier E (2025) "*Streptococcus pyogenes* EVs induce the alternative inflammasome via caspase-4/-5 in human monocytes" **EMBO Rep**, 26(19), 4847–4885.
11. Ajay Castro S, Passmore IJ, Ndeh D, Shaw HA, Ruda A, Burns K, Thomson S, Nagar R, **Alagesan K**, Reglinski M, Lucas K, Abouelhadid S, Schwarz-Linek U, Mawas F, Widmalm G, Wren BW and Dorfmueller HC (2025) "Recombinant production platform for Group A Streptococcus glycoconjugate vaccines" ***npj Vaccines*****,** 10(1), 16.

AWARDS – SELECTED LIST

1. **ECR Travel Award (2019)** – To attend Human Proteome Organization Conference (HUPO) 2019 in Adelaide, Australia.
2. **IBCarb trainee award (2015)** - To participate in COSTCM 1102/IBCarb Synthesis for nano- and glyco-sciences Spring Training School held at Bangor, United Kingdom.
3. **6th Journal of Mass Spectrometry Award (2014)** - Awarded at the 20th International Mass Spectrometry Conference (IMSC) in Geneva.
4. **Best poster award (2013)** - Awarded at the Joint Glycobiology meeting, Wittenberg, Germany.
5. Selected as one of the **best annotators** of the Mycobacterium tuberculosis glycogenome from over 800 researchers (**2010**).

SELECTED INVITED AND AWARD LECTURES

1. **Invited Speaker: MPUSP-UCMR Mini-Symposium, 2026, Germany.** 'The molecular playbook of bacterial survival: decoding proteome and glycome remodeling during stress and infection'
2. **Invited Speaker: AICTE Training and Learning Academy on 'Emerging trends in additive manufacturing for healthcare applications 06 Jan 2026 (Virtual).** 'From Amino to Glyco: The sweet side of proteomics' - *Invited by Dr. Preethi S, Sathyabama Institute of Science and Technology, Chennai, India.*
3. **Invited Speaker: IZW Seminar series 2024.** 'Sugar, Spice & Spectral Insights: (Glyco)Proteomics Unwrapped' - *Invited by Dr. Kristin Mühldorfer, Leibniz Institute for Zoo and Wildlife Research (IZW) in the Forschungsverbund Berlin e.V.*
4. **Invited Speaker: Technische Universität Berlin seminar series 2021 (Virtual)**. 'The sweet side of proteomics' - *Invited by Prof. Juri Rappsilber*
5. **Invited Speaker: ASBMB/EB Glycobiology Interest Group 2021 (Virtual).** 'Improved glycopeptide identification using High-field Asymmetric Waveform ion Mobility Spectrometry (FAIMS)' - *Invited by Dr. Nadine Samara & Prof. Amanda Lewis*
6. **Invited Speaker: CMB seminar 2020** at VIB Center for Medical Biotechnology, UGent Department of Biomolecular Medicine, Belgium. 'The Mass Spectrometry toolkit for clinical glycomics and glycoproteomics' – *Invited by Prof. Nico Callewaert*
7. **Award talk: 20th International Mass Spectrometry Conference (Journal of Mass Spectrometry Award symposium), 2014, Switzerland.** 'A novel, ultrasensitive approach for quantitative carbohydrate composition and linkage analysis using LC-ESI ion trap tandem mass spectrometry'

SELECTED POSTER PRESENTATION

1. **25th Human Proteome Organization World Congress (HUPO 2026), Singapore.** 'Rebel, Rebel: Breaking Norms with Protease Multiplexing for Glycoproteomics'
2. **2025 Gordon Research Conference – Glycobiology.** 'Rebel, Rebel: Breaking the norms of protease multiplexing'
3. **Joint Warren and Beilstein Symposium on Glycosciences 2021. Virtual.** 'Improved glycopeptide identification using High-field Asymmetric Waveform ion Mobility Spectrometry (FAIMS)'
4. **GlycoT 2020, Boston MA/Virtual.** 'Better together: relative retention time + spectral matching improves automated glycan identification'
5. **HUPO 2019, Australia**. 'SynGp one-stop-shop: synthetic glycopeptides assisted glycoproteomics'

EDUCATION

Doctor of Natural Sciences (Dr. rer. nat.) **Sep 2012 – Dec 2016**

Department of Biomolecular Systems,
Max Planck Institute of Colloids and Interfaces, Potsdam, Germany
Freie Universität Berlin, Institute of Chemistry and Biochemistry,
Department of Biology, Chemistry, and Pharmacy, Berlin, Germany

***Thesis**: "The Mass Spectrometry Toolkit for Glycoprotein Characterisation: Development of Novel Analytical Methods and Technologies for Glycomics and Glycoproteomics"*

- Established a simple approach for isolating glycosylated amino precursors required for solid-phase glycopeptide synthesis.
- Developed a cost-effective and robust technique for glycopeptide enrichment.
- Developed and implemented a glycomics tool based on porous graphitized carbon (PGC) nLC-MS.
- Operated, troubleshot, and maintained Ultimate 3000 nLC and Amazon Speed ETD MS, Bruker MALDI TOF MS.

Master of Science (M.Sc.) in Biochemistry **Aug 2010 – Sep 2012**

Department of Biochemistry, University of Oulu, Oulu, Finland.

***Major:** Protein Science and Biotechnology*

***Thesis**: "Novel methods for monosaccharide identification and quantitation based on Liquid Chromatography Electrospray Ionisation (LC-ESI) ion trap mass spectrometry"*

Bachelor of Technology (B.Tech) in Biotechnology **2005 - 2009**

Kumaraguru College of Technology, Coimbatore, Tamil Nadu, India (Affiliated to Anna University)

***Thesis**: "Isolation and identification of α-glucosidase inhibitors from medicinal plants"*

MEMBERSHIP

1. Australasian Proteomics Society
2. ASMS (American Society of Mass Spectrometry)
3. HUPO (Human Proteome Organization)
4. ASBMB (Australian Society for Biochemistry and Molecular Biology)

PUBLICATIONS – FULL LIST

Total citations: 994 (05-Oct-2026) **h-index: 15 (Google Scholar)**

1. Yu Y, **Alagesan K**, Frahm D, Charpentier E, Iatsenko I (2026) "Microbiota-induced fatty acid synthesis facilitates intestinal infection and immune-mediated damage in *Drosophila*" **bioRxiv** https://doi.org/10.64898/2026.04.14.718535
2. Rubinić M, Yu Y, **Alagesan K,** and Iatsenko I (2026) "A Pore-Forming Toxin Monalysin Contributes to Infection-Induced Suppression of Defecation in Female Drosophila" **PLOS Pathog. Accepted.** bioRxiv. https://doi.org/10.64898/2025.12.03.692093
3. Giralt-Zúñiga MJ, Jahn M, Franklin JL, **Alagesan K**, Kondrot F, Kaganovitch E, Hallenga L, Derado S, Hughes KT, Popp PF, Charpentier E, Dufour YS, Erhardt M (2026) "The cost-benefit trade-off of peritrichous flagellation in bacteria" **bioRxiv**. https://doi.org/10.64898/2026.08.20.746045
4. Krause K, Franch Arroyo S, Ugolini M, Kueck T, Sullivan TJ, Gálvez EJC, Muenzner M, Goosmann C, Brinkmann V, Frese CK, **Alagesan K**, Vierbuchen T, Heine H, Resch U, Sander LE and Charpentier E (2025) "*Streptococcus pyogenes* EVs induce the alternative inflammasome via caspase-4/-5 in human monocytes" **EMBO Rep**, 26(19), 4847–4885. https://doi.org/10.1038/s44319-025-00558-7
5. Rubinić M, Arias-Rojas A, Martinez KA, Klimek W, Paczia N, **Alagesan K**, Duneau D and Iatsenko I (2026) "Sex differences in *Drosophila* intestinal metabolism contribute to sexually dimorphic infection outcome and alter gut pathogen virulence"**.** ***Proc. Natl. Acad. Sci. U.S.A.*** 123 (26). https://doi.org/10.1073/pnas.2514992123
6. Cornejo FA, Driller K, Ahmed-Begrich R, Schmidt K, Jahn M, Shanmuganathan V, Hahnke K, Kondrot F, Wulff TF, Rämisch S, **Alagesan K**, Charpentier E, Turgay K (2026) "Stress-induced ribosome degradation in *Bacillus subtilis* is mediated by the RNase Y-specificity complex" ***Nat. Commun***. 17(1):4886. https://doi.org/10.1038/s41467-026-73310-x
7. Ajay Castro S, Passmore IJ, Ndeh D, Shaw HA, Ruda A, Burns K, Thomson S, Nagar R, **Alagesan K**, Reglinski M, Lucas K, Abouelhadid S, Schwarz-Linek U, Mawas F, Widmalm G, Wren BW and Dorfmueller HC (2025) "Recombinant production platform for Group A Streptococcus glycoconjugate vaccines" ***npj Vaccines*****,** 10(1), 16. https://doi.org/10.1038/s41541-025-01068-2
8. Ignatov D, Shanmuganathan V, Ahmed-Begrich R, **Alagesan K**, Hanke K, Wang C, Krause K, Cornejo FA, Funke K, Erhardt M, Frese CK, and Charpentier E (2025) "RNA-binding protein YebC enhances translation of proline-rich amino acid stretches in bacteria". ***Nat Commun*** 16, 6262. https://doi.org/10.1038/s41467-025-60687-4
9. Alagesan K and Charpentier E (2023) "Systems-Wide Site-Specific Analysis of Glycoproteins" In K. Gevaert (Ed.), ***Mass Spectrometry-Based Proteomics*****,** 2718, 151–165. New York, NY: Springer US. https://doi.org/10.1007/978-1-0716-3457-8_9
10. **Alagesan K**, Ahmed-Begrich R and Charpentier E (2022) "Improved N- and O-Glycopeptide Identification using High-Field Asymmetric Waveform Ion Mobility Spectrometry (FAIMS) (preprint)." **Biochemistry**. https://doi.org/10.1101/2022.12.12.520086
11. **Alagesan K**, Schirmeister F, Möginger U, Everest-Dass A, Altmann F, Seeberger PH, Von Itzstein M, Packer NH and Kolarich D (2021**)** "Better together – relative retention time plus spectral matching improves automated glycan characterization using PGC-nLC-IT-ESI-MS/MS (preprint)." Biochemistry. https://doi.org/10.1101/2021.08.04.451140
12. Kawahara R, Chernykh A, **Alagesan K**, Bern M, Cao W, Chalkley RJ, Cheng K, Choo MS, Edwards N, Goldman R, Hoffmann M, Hu Y, Huang Y, Kim JY, Kletter D, Liquet B, Liu M, … Thaysen-Andersen

M (2021) “Community evaluation of glycoproteomics informatics solutions reveals high-performance search strategies for serum glycopeptide analysis” ***Nat Methods*****,** 18(11), 1304–1316. https://doi.org/10.1038/s41592-021-01309-x

13. Lisacek F, **Alagesan K**, Hayes C, Lippold S and De Haan N (2021) “Bioinformatics in Immunoglobulin Glycosylation Analysis” In M. Pezer (Ed.), ***Antibody Glycosylation***, 112, 205–233. Cham: Springer International Publishing. https://doi.org/10.1007/978-3-030-76912-3_6
14. Oliveira T, Zhang M, Joo EJ, Abdel-Azim H, Chen C-W, Yang L, Chou C-H, Qin X, Chen J, **Alagesan K**, Almeida A, Jacob F, Packer NH, Von Itzstein M, Heisterkamp N and Kolarich D (2021) “Glycoproteome remodeling in MLL-rearranged B-cell precursor acute lymphoblastic leukemia” ***Theranostics***, 11(19), 9519–9537. https://doi.org/10.7150/thno.65398
15. **Alagesan K**, Hoffmann M, Rapp E and Kolarich D (2020) “Glycoproteomics Technologies in Glycobiotechnology” In E. Rapp and U. Reichl (Eds.), ***Advances in Glycobiotechnology***, 175, 413–434. Cham: Springer International Publishing. https://doi.org/10.1007/10_2020_144
16. De Leoz MLA, Duewer DL, Fung A, Liu L, Yau HK, Potter O, Staples GO, Furuki K, Frenkel R, Hu Y, Sosic Z, Zhang P, Altmann F, Grunwald-Grube C, Shao C, Zaia J, Evers W, … Stein SE (2020) “NIST Interlaboratory Study on Glycosylation Analysis of Monoclonal Antibodies: Comparison of Results from Diverse Analytical Methods” ***Molecular & Cellular Proteomics*****,** 19(1), 11–30. https://doi.org/10.1074/mcp.RA119.001677
17. Pralow A, Cajic S, **Alagesan K**, Kolarich D and Rapp E (2020) “State-of-the-Art Glycomics Technologies in Glycobiotechnology” In E. Rapp and U. Reichl (Eds.), ***Advances in Glycobiotechnology***, 175, 379–411. Cham: Springer International Publishing. https://doi.org/10.1007/10_2020_143
18. **Alagesan K**, Hinneburg H, Seeberger PH, Silva DV and Kolarich D (2019) “Glycan size and attachment site location affect electron transfer dissociation (ETD) fragmentation and automated glycopeptide identification” ***Glycoconj J*****,** 36, 487–493. https://doi.org/10.1007/s10719-019-09888-w
19. **Alagesan K** and Kolarich D (2019) “To enrich or not to enrich: Enhancing (glyco)peptide ionization using the CaptiveSpray nanoBooster™ **(**preprint).” Biochemistry. https://doi.org/10.1101/597922
20. **Alagesan K**, Silva DV, Seeberger PH and Kolarich D (2019) “A novel, ultrasensitive approach for quantitative carbohydrate composition and linkage analysis using LC-ESI ion trap tandem mass spectrometry (preprint).” Biochemistry. https://doi.org/10.1101/853036
21. **Alagesan K**, Everest-Dass A and Kolarich D (2018) “Isomeric Separation and Characterisation of Glycoconjugates” In Y. Yamaguchi and K. Kato (Eds.), ***Glycobiophysics***, 1104, 77–99. Singapore: Springer Singapore. https://doi.org/10.1007/978-981-13-2158-0_5
22. **Alagesan K**, Khilji SK and Kolarich D (2017) “It is all about the solvent: on the importance of the mobile phase for ZIC-HILIC glycopeptide enrichment” ***Anal Bioanal Chem***, 409(2), 529–538. https://doi.org/10.1007/s00216-016-0051-6
23. O’Neill E, Kuhaudomlarp S, Rejzek M, Fangel J, **Alagesan K**, Kolarich D, Willats W and Field R (2017) “Exploring the Glycans of Euglena gracilis” ***Biology***, 6(4), 45. https://doi.org/10.3390/biology6040045
24. Kolarich D, Windwarder M, **Alagesan K** and Altmann F (2015) “Isomer-Specific Analysis of Released N-Glycans by LC-ESI MS/MS with Porous Graphitized Carbon” In A. Castilho (Ed.), ***Glyco-Engineering*****,** 1321, 427–435. New York, NY: Springer New York. https://doi.org/10.1007/978-1-4939-2760-9_29
25. Campbell MP, Nguyen-Khuong T, Hayes CA, Flowers SA, **Alagesan K**, Kolarich D, Packer NH and Karlsson NG (2014) “Validation of the curation pipeline of UniCarb-DB: Building a global glycan

reference MS/MS repository" ***Biochimica et Biophysica Acta (BBA) - Proteins and Proteomics***, 1844(1), 108–116. https://doi.org/10.1016/j.bbapap.2013.04.018

26. Klatt S, Rohe M, **Alagesan K** Kolarich D, Konthur Z and Hartl D (2013) "Production of Glycosylated Soluble Amyloid Precursor Protein Alpha (sAPPalpha) in *Leishmania tarentolae*" ***J. Proteome Res***., 12(1), 396–403. https://doi.org/10.1021/pr300693f
27. Williams CF, Lloyd D, Kolarich D, **Alagesan K**, Duchêne M, Cable J, Williams D and Leitsch D (2012) "Disrupted intracellular redox balance of the diplomonad fish parasite Spironucleus vortens by 5-nitroimidazoles and garlic-derived compounds" ***Veterinary Parasitology***, 190(1–2), 62–73. https://doi.org/10.1016/j.vetpar.2012.05.011